SS-31 (Mitochondrial Peptide)
What It Is

SS-31 is a synthetic peptide made of four amino acids that targets the inner membrane of your mitochondria, which is where your body produces energy. Its technical name is elamipretide, and it was developed by Dr. Hazel Szeto at Weill Cornell Medical College and Dr. Peter Schiller at the Montreal Clinical Research Institute. The name SS-31 stands for Szeto-Schiller peptide 31.

On September 19, 2025, elamipretide received accelerated approval from the FDA under the brand name Forzinity for treating Barth syndrome, an extremely rare genetic disorder affecting about 150 people in the United States. This makes SS-31 the first FDA approved treatment that specifically targets mitochondria, and the first mitochondria-targeted peptide to receive FDA approval for any indication.

What makes SS-31 fundamentally different from other mitochondrial compounds is what it does. NAD+ provides fuel. MOTS-c activates pathways. SS-31 repairs the structural damage that prevents your mitochondria from working properly in the first place. If MOTS-c is the supercharger and NAD+ is the fuel, SS-31 is the mechanic. It is fixing structural damage, cracks, leaks, broken components. Using fuel and upgrades on a damaged engine gives you limited results. You have to fix the damage first.

This is exactly why I say for folks over 35 to 40 who have never done peptides before and are starting on their health journey, absolutely start with SS-31 first to repair the problem before trying to optimize it.

How It Works

Before understanding SS-31, you need to understand how your body actually makes energy, because if you do not understand this, the rest will not make any sense.

Energy Production Basics

Your mitochondria are the power plants inside your cells. They take the food you eat and convert it into usable energy called ATP. ATP is the energy currency your body uses for everything from muscle contractions to brain function to tissue repair to simply staying alive.

Your body stores almost zero ATP. You only have about 100 grams on hand at any given time, which is enough for maybe a few seconds of intense activity. So your body has to constantly remake ATP all day, every day just to keep you functioning. A moderately active person cycles through their entire body weight in ATP daily.

About 90% of your ATP gets made through the electron transport chain. This happens on the inner membrane of your mitochondria. You have a series of protein complexes lined up on the inner membrane, and electrons get passed from one complex to the next. As those electrons move through, they pump protons across the membrane and that creates a gradient. That gradient is what powers an enzyme called ATP synthase, which actually makes the ATP.

The key here is efficiency. When this system works properly, electrons flow smoothly through the chain and you get maximum ATP with minimal waste. But electrons can leak out during this process. When electrons leak, they react with oxygen and create reactive oxygen species, which are essentially free radicals. Those free radicals cause damage to your cells. As you age, this leakage increases. More electrons leak, more free radicals get produced, and those free radicals start damaging the very structures that are supposed to be producing your energy.

Cardiolipin: The Critical Component

There is a specific component of the inner mitochondrial membrane that is critical for keeping this whole system organized and efficient. When it gets damaged, everything falls apart. That component is called cardiolipin.

Cardiolipin is a type of fat molecule, and it only exists in one place in your entire body: the inner membrane of your mitochondria where energy production happens.

Cardiolipin does three critical things. First, it provides structural support for the cristae, which are the folds inside your mitochondria that increase the surface area for energy production. More surface area means more space to make ATP. Second, it keeps all those protein complexes in the electron transport chain organized and lined up properly in structures called supercomplexes. When cardiolipin is healthy, electrons flow smoothly from one complex to the next with minimal leakage. Third, it stabilizes the interaction between cardiolipin and cytochrome c, improving electron transfer efficiency.

The free radicals produced by electron leakage do not just float around creating random damage. They specifically target and damage cardiolipin. And when cardiolipin gets damaged, the protein complexes that were lined up nicely start falling out of position. Electrons cannot flow smoothly anymore. More electrons leak out. Those electrons create more free radicals. And those free radicals damage more cardiolipin. Now you are in a vicious cycle.

The result is your ATP production drops, your energy goes down, oxidative stress goes up, and the damage compounds over time.

How SS-31 Fixes This

SS-31 has a positive electrical charge and cardiolipin has a negative electrical charge. So SS-31 is attracted to cardiolipin like a magnet. It concentrates in your mitochondria at levels 5000 times higher than in the surrounding cytoplasm within 15 minutes of administration.

Once it binds to cardiolipin, SS-31 stabilizes the membrane structure. It prevents cardiolipin from getting damaged by free radicals. It keeps all those protein complexes lined up properly. And it preserves the structure of the cristae that give you more surface area for energy production.

The result is more efficient electron transfer, less electron leakage (studies show 40 to 60% reduction in free radical production), improved ATP production, and protected mitochondrial structure.

SS-31 does not just clean up free radicals after they have already been created like traditional antioxidants do. Traditional antioxidants are like street sweepers, cleaning up pollution after it is already in the air. SS-31 fixes the source of the leak that is creating the pollution in the first place.

Benefits

Mitochondrial Repair

The primary benefit is restoration of damaged mitochondrial function. In aged mice (equivalent to humans in their late 70s), 8 weeks of SS-31 treatment improved mitochondrial morphology, restored cristae structure, and reduced markers of cellular senescence. This was demonstrated in kidney glomerular cells, cardiac tissue, and skeletal muscle. Importantly, SS-31 did not create more mitochondria. It made the existing mitochondria work better. SS-31 is about quality, not quantity.

Improved Energy Production

By stabilizing the electron transport chain and reducing electron leakage, SS-31 increases ATP output from existing mitochondria. Clinical trials in Barth syndrome patients showed significant improvements in the 6 Minute Walk Test (96 meters of improvement), indicating better exercise capacity and energy availability.

Reduced Oxidative Stress

SS-31 decreases mitochondrial reactive oxygen species production at the source. Studies show 40 to 60% reduction in ROS production when cardiolipin is stabilized. This is fundamentally more effective than scavenging ROS after they form.

Organ Protection

Research demonstrates protective effects across multiple organ systems including heart, kidney, brain, and skeletal muscle. This makes sense because all these tissues depend heavily on mitochondrial function. The heart and kidneys have particularly high mitochondrial density and are especially vulnerable to mitochondrial dysfunction.

Functional Improvement in Aging

A 2025 study in Aging Cell showed that 8 weeks of SS-31 treatment in aged mice mitigated frailty accumulation and partially reversed age-related declines in cardiac strain and skeletal muscle fatigue resistance. Pathway analysis revealed pro-longevity shifts including upregulation of fatty acid metabolism, mitochondrial translation, and oxidative phosphorylation genes, alongside reduced inflammation markers.

What to Expect

Timeline

SS-31 reaches peak concentration in your blood within 15 minutes of administration. It has a half-life of about 2 hours, but it accumulates in your mitochondria and stays there much longer than blood levels would suggest.

Most people begin to notice changes within the first 1 to 2 weeks, typically starting with subtle improvements in energy and recovery. The full repair phase plays out over 4 to 8 weeks.

Who Benefits Most

SS-31 makes the most sense for adults over 35 to 40 who are experiencing energy decline, slow recovery, and metabolic issues. By the time you are in your 40s with these symptoms, you likely have 10 to 20 years of accumulated mitochondrial damage. Damaged mitochondria cannot respond effectively to optimization signals. This is why some people supplement with NAD+ and do not feel anything. The machinery is too broken to use the fuel.

For younger or metabolically healthy individuals, SS-31 may not even be necessary. But if you are 35 plus with a lifetime of habits stacked up, repair first.

The Connection to Other Compounds

Barth syndrome patients have damaged cardiolipin from a genetic defect they were born with. If you are in your 40s or older, you have damaged cardiolipin from 20, 30, or 40 years of accumulated stress, inflammation, and oxidative damage. Different cause, same downstream problem, same solution.

The Foundation Matters

SS-31 is a tool that enhances the work you are already doing. Training, nutrition, and sleep are always going to remain your primary focus. If you are not building a foundation of healthy, positive habits, then adding any peptides, supplements, or anything in between is not going to help you. It is just going to put a band-aid on a problem that you are going to continue to create. Fix the foundation first.

What the Science Shows

SS-31 has extensive preclinical data and has been tested in 18 human clinical trials across multiple conditions. It has the broadest evidence base of any mitochondria-targeted peptide.

Barth Syndrome Trials: TAZPOWER (Phase 2/3, Human)

The TAZPOWER trial tested 40 mg daily of elamipretide via subcutaneous injection in 12 patients with Barth syndrome. While the initial 12-week crossover phase did not meet primary endpoints, the 168-week open-label extension (published in Genetics in Medicine, 2024) showed sustained long-term improvements:

- 6 Minute Walk Test improved by 96.1 meters ($P = .003$)
- Leg muscle strength improved by over 45%
- Heart function improved by 45%
- Barth Syndrome Symptom Assessment improved by 2.1 points
- 3D left ventricular stroke, end-diastolic, and end-systolic volumes improved
- Cardiolipin levels improved compared to baseline

Ten patients entered the open-label extension with 8 reaching the week 168 visit. Elamipretide was well tolerated throughout, with injection-site reactions being the most common adverse events.

Aging and Kidney Function (Preclinical, 2017)

Sweetwyne and colleagues studied mice aged 24 to 26 months (equivalent to humans aged 70 to 90). Eight weeks of SS-31 treatment reversed age-related decline in ATP production, restored the balance between free radical production and the body's ability to neutralize them, improved mitochondrial morphology in glomerular cells, reduced expression of senescence markers p16 and senescence-associated beta-galactosidase, increased parietal epithelial cell density, reduced glomerulosclerosis, and preserved glomerular endothelial density. Published in Kidney International.

Aging Cardiac and Skeletal Muscle (Preclinical, 2025)

Mitchell, Pharaoh, and colleagues (Harvard Medical School and University of Washington) published a study in Aging Cell showing that 8 weeks of elamipretide treatment in aged C57BL/6J mice mitigated frailty accumulation and partially reversed age-related cardiac dysfunction and skeletal muscle weakness. Pathway analyses revealed pro-longevity shifts including upregulation of fatty acid metabolism, mitochondrial translation, and oxidative phosphorylation genes. Interestingly, despite the functional improvements, the researchers did not detect statistically significant changes in epigenetic or transcriptomic age, suggesting that functional improvement can occur independently of molecular age reversal.

Barth Syndrome Cardiac Morphology (Preclinical, 2024)

Machiraju and colleagues showed in a murine model that SS-31 treatment ameliorated cardiac mitochondrial morphology and defective mitophagy, providing mechanistic support for the clinical findings. Published in Scientific Reports.

Exercise Tolerance in Aged Mice (Preclinical, 2019)

Siegel and colleagues at the University of Washington showed that SS-31 treatment in aged mice improved redox balance and reversed age-related exercise intolerance. The compound increased muscle mass and endurance and improved kidney function. Published in Free Radical Biology and Medicine.

Heart Failure Studies (Human)

Clinical trials in heart failure with preserved ejection fraction showed mixed results. Some cardiac parameters improved, but primary endpoints were not consistently met. Research continues in this area.

Mitochondrial Myopathy (Human)

Trials in primary mitochondrial myopathy showed improvements in some secondary endpoints including reduced fatigue and muscle complaints compared to placebo, but did not meet primary efficacy endpoints. The compound was generally well tolerated.

What People Report

The following is aggregated from external platforms including Reddit, peptide forums, and community conversations. This is anecdotal data and does not carry the same weight as published research.

Energy and Recovery

Users commonly report improved baseline energy levels and faster recovery from training, typically noticing changes within the first 1 to 2 weeks. Reports describe the improvement as a more stable, sustained energy rather than a stimulant-like burst.

Injection Site Reactions

This is the most frequently discussed side effect in user communities. Redness, itching, and occasional welts at the injection site are common. Most users manage this with antihistamines or hydrocortisone cream and report that reactions diminish over time. Rotating injection sites helps.

Subtle Effects

Some users express frustration that the effects are not dramatically noticeable. This makes sense given that SS-31 is repairing structural damage rather than providing an acute performance boost. The benefits are often described as "things working better" rather than a distinct feeling.

Pairing with NAD+ and MOTS-c

Users who follow the repair-then-optimize sequence (SS-31 first, then MOTS-c) frequently report that they noticed more benefit from MOTS-c and NAD+ after completing an SS-31 cycle, consistent with the idea that repairing the machinery first allows other compounds to work more effectively.

Dosing Protocol

Clinical Doses

The Barth syndrome trials used 40 mg daily subcutaneously. Those patients have severe mitochondrial dysfunction from a genetic condition. That dosing is for people with fundamentally broken mitochondria, not age-related decline.

The mouse studies used doses that translate to roughly 5 to 7 mg daily in humans.

Practical Protocol for Age-Related Decline

For otherwise healthy individuals experiencing age-related mitochondrial decline:

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REPAIR PROTOCOL
Dose:          1 to 2 mg per day
Frequency:     Once daily
Timing:        No fasting required, no specific timing constraints
Administration: Subcutaneous injection
Duration:      4 to 8 weeks (repair phase)

RECONSTITUTION (10 mg vial)
Bac water:     2 mL
Concentration: 500 mcg per 10 units on insulin syringe
               1 mg per 20 units on insulin syringe
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Protocol Sequencing (Repair First Approach)

For people over 35 to 55 with declining energy, slower recovery, and metabolic issues:

Weeks 1 through 4 (or up to 8): Repair Phase SS-31 at 1 to 2 mg daily plus NAD+ at 100 mg subcutaneously three times per week. You are repairing accumulated mitochondrial damage while giving your cells the extra fuel they need to optimize energy.

Weeks 5 through 12: Optimize Phase Replace SS-31 with MOTS-c. Start with 10 mg split into three injections per week (Monday, Wednesday, Friday). Continue NAD+ at 100 mg three times per week. Now you are optimizing the repaired mitochondria.

After the 12-week cycle: Take 4 to 8 weeks completely off. Then you can repeat this protocol 2 to 3 times per year.

Prevention First Protocol

For younger or healthier individuals who want preventative optimization:

Start with MOTS-c at 10 mg split into three injections per week (Monday, Wednesday, Friday) plus NAD+ at 100 mg three times per week. At weeks 4 through 8, assess whether you still have significant fatigue, poor recovery, or metabolic issues. If yes, add SS-31 at 1 to 2 mg daily for 4 to 8 weeks while continuing MOTS-c and NAD+. If not, continue with just MOTS-c and NAD+ for maintenance.

A Note on Higher Doses

You may see dosages recommended online as high as 5 mg, 10 mg, or even 50 mg per day. Those protocols are for people with diseases like Barth syndrome or other conditions involving systemic mitochondrial dysfunction. If you are in your 40s with declining energy and slow recovery, you do not have Barth syndrome. You have accumulated damage from decades of life. Lower doses work for repair because your mitochondria are damaged, not fundamentally broken.

Stacking Context

SS-31 + NAD+

These work together directly. SS-31 repairs the mitochondrial structure. NAD+ provides the fuel that repaired mitochondria can now use efficiently. If your mitochondria are damaged, just adding more NAD+ might not work because converting NAD+ precursors into usable NAD+ requires ATP, and damaged mitochondria cannot produce ATP efficiently. Fix the machinery first with SS-31, then fuel it with NAD+.

SS-31 + MOTS-c (Sequential or Concurrent)

The original recommendation was always SS-31 first for repair, then MOTS-c for optimization. That is still the right call if you have significant mitochondrial dysfunction. If the engine is blown, you fix it before you upgrade it.

But a 2026 study by Gudiksen and colleagues changed the picture. They showed that MOTS-c improves intrinsic mitochondrial quality and reduces ROS through a completely different mechanism than SS-31. SS-31 reduces ROS at the membrane level by preventing electron leakage through cardiolipin stabilization. MOTS-c reduces ROS at the nuclear level by upregulating antioxidant enzymes through AMPK, PGC-1 alpha, and NRF2 pathways. Different locations, different dependencies, no overlap.

That means for people with moderate age-related decline who are otherwise active and functional, running both concurrently from day one lets you hit structural repair and functional optimization at the same time. And there is preclinical data suggesting MOTS-c may support cardiolipin biosynthesis through AMPK activation, meaning it could help produce more of the molecule SS-31 is protecting.

For significant dysfunction: sequence SS-31 first for 4 to 8 weeks, then add MOTS-c. For moderate decline: concurrent use is now a supported option.

SS-31 + Epithalon + FOXO4-DRI (Longevity Stack for 50+)

The mitochondrial repair stack for people over 50. FOXO4-DRI eliminates senescent cells, Epithalon repairs telomeres, and SS-31 repairs mitochondria. These can all be run together. After SS-31 is complete, transition to MOTS-c and NAD+ for optimization.

SS-31 + GLP-1 Agonists

No interaction concerns. Completely different mechanisms. Can be run concurrently without timing conflicts.

SS-31 + Growth Hormone Peptides

No interaction concerns. Different timing requirements. GH peptides require fasting. SS-31 does not. Inject SS-31 at any time and keep GH peptides on their own fasting schedule.

SS-31 + BPC-157 or TB-500

No interaction concerns. Can be run concurrently. Different targets entirely.

Common Questions

Why should I start with SS-31 instead of MOTS-c? If you have significant mitochondrial dysfunction, SS-31 first is still the right call. MOTS-c tells your cells to produce more energy (it is the supercharger). SS-31 repairs the structural damage (it is the mechanic). Revving a broken engine harder just creates more exhaust. Fix the structure first, then optimize. But if you have moderate age-related decline and are otherwise active and functional, newer research shows these two compounds reduce ROS through completely non-overlapping mechanisms, membrane level for SS-31 and nuclear transcription level for MOTS-c, which means running them concurrently is a viable option. For younger, metabolically healthy people, starting with MOTS-c alone may be appropriate because they may not have significant structural damage yet.

How quickly does SS-31 work? It reaches peak concentration within 15 minutes and has a half-life of about 2 hours. But mitochondrial repair takes time. Most people notice subtle energy and recovery improvements within 1 to 2 weeks, with the full repair phase playing out over 4 to 8 weeks.

Why are the injection site reactions so common? SS-31 can trigger mild histamine responses at the injection site. This is the most frequently reported side effect across all clinical trials. Rotating injection sites helps, and most reactions are mild. Antihistamines or hydrocortisone cream can manage them if needed.

Can I just take NAD+ without SS-31? You can, but if your mitochondria are damaged, you may not get much benefit. Converting NAD+ precursors into usable NAD+ requires ATP. If your mitochondria cannot produce ATP efficiently, they cannot use NAD+ precursors. This is exactly why some people supplement with NAD+ and do not feel anything. The machinery is too broken to use the fuel.

Is SS-31 cleared through the liver? No. SS-31 is completely cleared through the kidneys with no processing through the liver. If you have kidney issues, talk to your doctor before considering this.

Side Effects

In Published Research

Across all clinical trials (over 3 years of daily use in Barth syndrome patients), the safety profile is strong. No serious drug-related adverse events have been consistently reported.

Common:

- Erythema (redness) at injection site: 57%
- Pruritus (itching): 47%
- Pain at injection site: 20%
- Urticaria (hives): 20%
- Irritation: 10%

Less Common:

- Headache (reported in some trials)
- Fatigue (paradoxically, some users report initial fatigue before improvement)
- Gastrointestinal discomfort (rare)

Most injection site reactions are mild and resolve without intervention. Rotating injection sites helps reduce local reactions.

What Users Report

User reports align closely with clinical trial data. Injection site reactions are the primary complaint, with most users describing them as manageable. Some users report initial fatigue in the first few days that resolves as the repair process progresses.

Contraindications and Precautions

Do Not Use If You Have:

- Severe kidney disease (SS-31 is entirely renally excreted)
- Known hypersensitivity to the peptide components
- Pregnancy or breastfeeding (insufficient safety data)

Use Caution With:

- Mild to moderate kidney impairment (consult your doctor)
- Individuals on medications affecting mitochondrial function
- Autoimmune conditions (mitochondrial support may affect immune cell function)
- Heart conditions (consult physician, as some trials showed variable cardiac effects)

Drug Interactions:

- No well-established drug interactions
- Theoretically, combining with other mitochondria-targeting compounds should be monitored

Pharmacokinetics:

- Peak plasma levels within 15 minutes of subcutaneous administration
- Half-life approximately 2 hours
- 100% renal clearance, no hepatic metabolism
- Accumulates in mitochondria at concentrations 5000 times higher than cytoplasm